

The Parallel Universe: parallel imaging and novel acquisition techniques

17.1 Introduction

We saw in chapter 11 that rapidly switched fields can lead to peripheral nerve stimulation. As figure 17.1 shows, in order to halve the scan time, the gradient slew rate must be quadrupled. Whilst producing higher and higher gradient slew rates is merely a matter for improved engineering, there comes a point when PNS becomes inevitable and thus the ultimate speed limit of MRI is physiologically determined. This imposes an apparently non-negotiable limit on the acquisition techniques we have encountered so far in this book. So does that mean fast imaging development has ground to a standstill? Surprisingly not, as even this fundamental limitation can be overcome, (or rather, circumvented) by using parallel imaging and reconstruction techniques together with phased array technology for even faster scanning. It needs to be stressed here that parallel imaging methods are not sequences, but entirely new ways of acquiring and reconstructing images. The two classic formulations of parallel imaging are SMASH and SENSE, working in k-space and image space respectively. This chapter will explain the various types of parallel imaging and other novel reconstruction and acquisition techniques, how they work, what advantages they provide and the image quality trade-offs involved.

In this chapter we will see that parallel imaging

- makes MR acquisitions faster by a factor known as the reduction factor R ,
- can be applied to any existing MR sequence, including EPI,
- is enabled by phased array technology and may be performed in k-space or in image space,

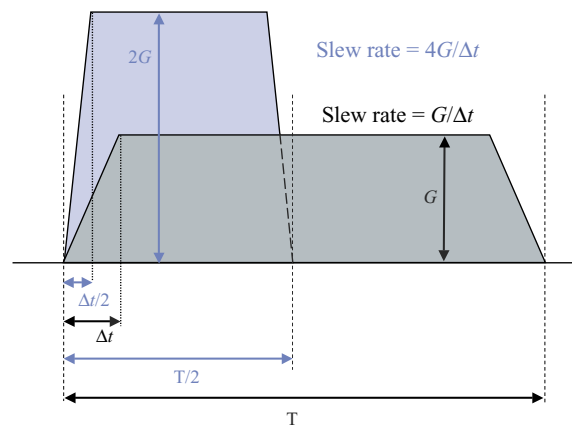


Figure 17.1 In order to halve the acquisition time whilst maintaining the same resolution and field of view, the gradient strength must be doubled and the gradient slew rate quadrupled.

- reduces the number of phase-encode steps you need to actually acquire an image,
- involves an image quality trade-off.

Additionally

- non-Cartesian acquisition methods can offer some advantages such as very short echo times, self-navigation and increased SNR.
- You will also learn lots of lovely new acronyms!

17.2 Groundwork

To understand parallel imaging we need to revisit two concepts considered earlier in the book, namely: array

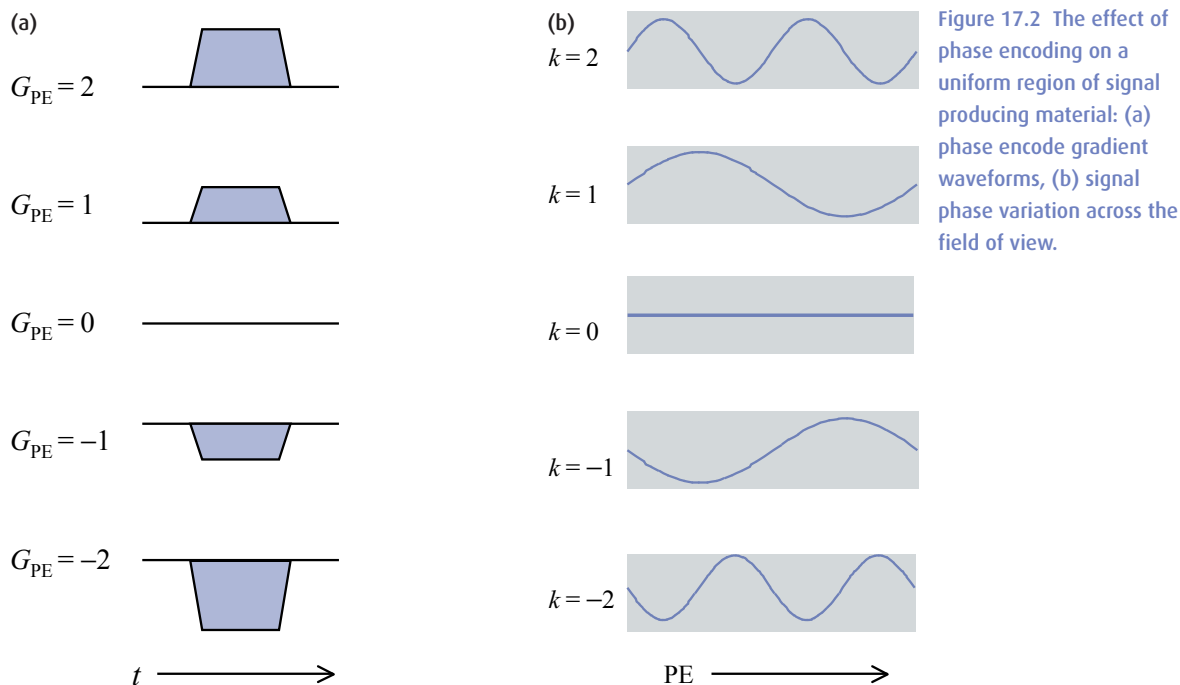


Figure 17.2 The effect of phase encoding on a uniform region of signal producing material: (a) phase encode gradient waveforms, (b) signal phase variation across the field of view.

coils (chapter 9) and phase encoding (chapter 7). If you are confident you understand both these you can skip forward to section 17.2.3.

17.2.1 Simple conceptual explanation of k-space and phase encoding

Phase encoding was explained in detail in chapter 7. For the present purpose we need to see that each phase-encode gradient step adds a further 2π of phase change across the image field of view (figure 17.2). Alternatively we can say that each line of k-space is separated by $2\pi/\text{FOV}$. Each phase encoding step sensitises the MR acquisition to specific patterns of signal distribution, or in other words, picks out a particular spatial frequency of the image. The steps of phase encoding and signal acquisition need to be repeated until k-space is filled, prior to Fourier transformation to generate the final image. If we double the separation between the lines of k-space by skipping lines of k-space we halve the image field of view and if the imaged object exceeds this, aliasing or foldover will occur.

17.2.2 Basic principles of phased arrays

The conventional use of phased array coils is to achieve superior SNR. Each array element is sensitive to a smaller volume of tissue than an equivalent larger coil. Inductively coupled noise, which usually dominates the noise formation process, has a theoretical strong dependence on the volume of tissue to which the coil is sensitive. The smaller array elements thus 'see' less noise than larger coils (figure 17.3). Each array element has a separate receiver channel, with up to 8 channels being common. Additionally arrays benefit from the combination of uncorrelated noise sources offering the same sort of advantage that quadrature coils have over linear coils (see section 9.4.3). Separate reconstructions are carried out for each array element with the final image being a combination of these. The increased SNR can be used to achieve higher resolution, shorter scan times, greater anatomical coverage or a combination of all three. Many clinical applications would be impossible without array coils. In parallel imaging we are going to trade part of the superior SNR to give us shorter acquisition times.

17.2.3 Coil sensitivity profiles

In the conventional use of phased arrays, the images produced from each element are combined to form the resultant image. The spatial variations inherent to the individual coil responses, or the coil sensitivities, have been combined. This is illustrated in figure 17.4 where individual sensitivity profiles are shown for two coils. The determination of array element sensitivity profiles or maps and the linear combination of signals from array elements are both crucial aspects of parallel imaging.

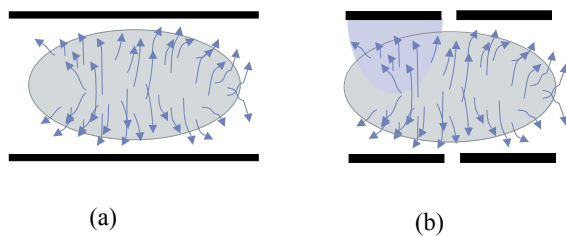


Figure 17.3 The inductive noise-reduction principle of phased array coils. (a) A larger single coil “sees” more noise from the greater volume of tissue it surrounds. (b) Smaller coil elements are less sensitive to inductive noise whilst adequate signal coverage is achieved by combining images from the separate elements.

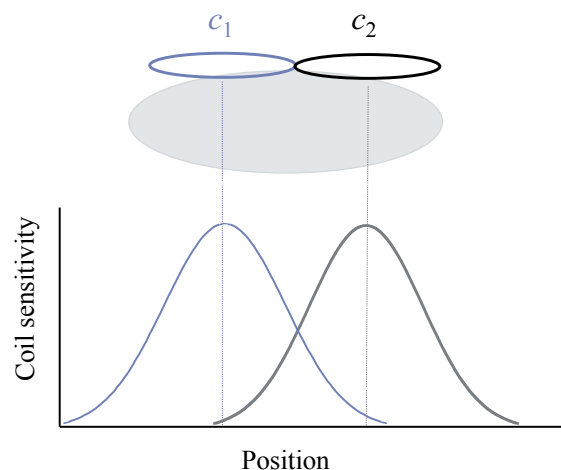


Figure 17.4 Coil sensitivity profiles: two loop surface coils.

17.3 Making SENSE: parallel imaging in image space

In parallel imaging we are going to speed up scanning by acquiring fewer lines of k-space and using our knowledge of the array coils used to unravel the aliased images or to generate the missing lines of k-space. The former technique employs parallel imaging reconstruction in image space, the latter in k-space. In both methods the scan time is reduced by a reduction factor R :

$$\text{Scan time}_{2D} = \frac{\text{NSA} \times \text{TR} \times N_{\text{PE}}}{R \times \text{ETL}}$$

where N_{PE} is the full (unreduced) PE matrix size and ETL the echo train length or turbo-factor (where appropriate) for 2D scans. For 3D scans

$$\text{Scan time}_{3D} = \frac{\text{NSA} \times \text{TR} \times N_{\text{PE}} \times N_{\text{SS}}}{R_{\text{PE}} \times R_{\text{SS}} \times \text{ETL}}$$

where R_{PE} and R_{SS} are reduction factors for each phase-encode direction.

17.3.1 SENSE

SENSE (SENSitivity Encoding) was the first parallel imaging technique to be realised commercially (by Philips). ASSET on General Electric scanners and SPEEDER (Toshiba) are broadly similar. In SENSE a reduced k-space is acquired by using fewer phase-encode gradient steps in conjunction with phased array coil acquisition.

SENSE reconstruction works post-Fourier transformation on the images. In a conventional acquisition if we omit every second phase-encode step we will halve the scan time, but will get an aliased image. In parts of the image aliased signal is superimposed upon unaliased signal. However the position in the resultant image of the aliased signal contribution is entirely predictable from a knowledge of the field of view. What is unknown is the intensity of the aliased component. The trick in SENSE is to apply knowledge of the sensitivities of the coil elements to calculate the aliased signal component at each point.

An example with a reduction factor of 2 and two coil elements is shown in figure 17.5 where the point P in the aliased image can be seen to be composed of signal

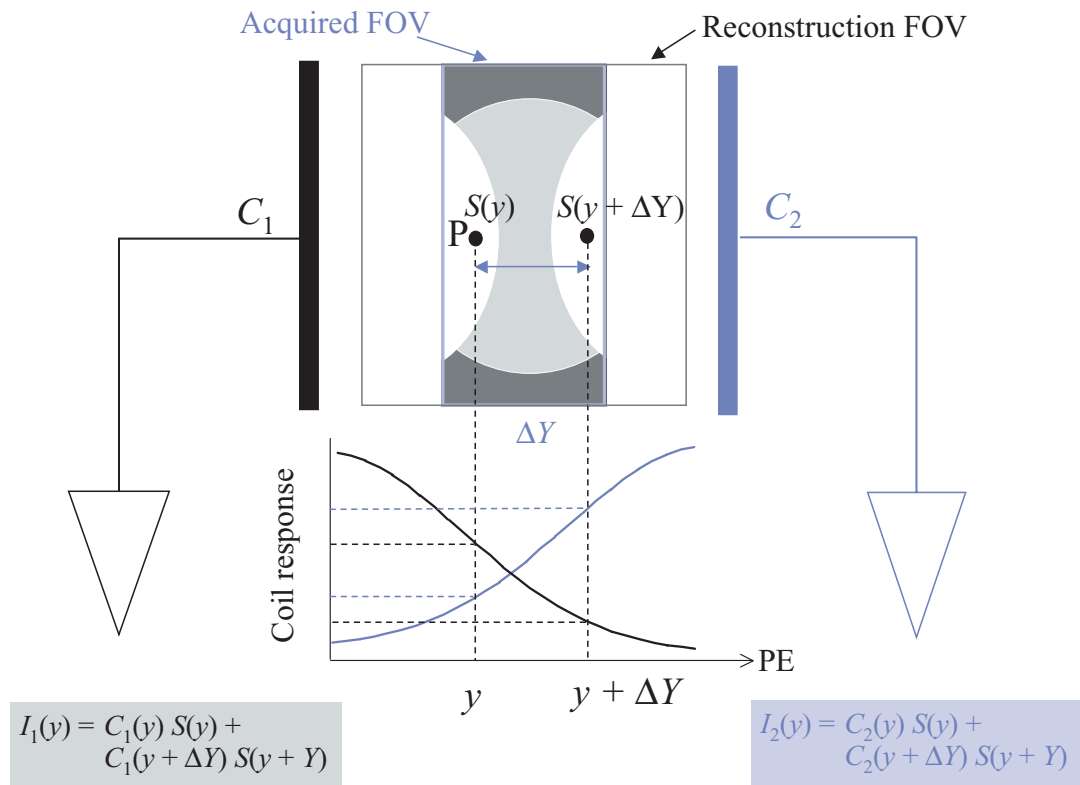


Figure 17.5 Image reconstruction in SENSE. Prior knowledge of the coil sensitivity profiles permits the unfolding of the image.

originating from two different locations: $S(y)$ from the true location y and $S(y + \Delta Y)$ from the aliased position. Provided we know the coil sensitivities we can calculate the true and the displaced signal components. Having computed the aliased signal component $S(y + \Delta Y)$, it can be reassigned to its proper location.

Figure 17.6 illustrates the acquisition and reconstruction with clinical images. Firstly sensitivity maps are generated for each coil element from a short, low resolution calibration scan, then an aliased image is reconstructed for each element before being fed into the SENSE reconstructor to produce the final image. The principle can be applied for a higher reduction factor R as shown in the box 'Making SENSE with maths'. For SENSE to work at all, there must be a coil sensitivity variation along the phase-encode direction. For array coils with appropriate geometry and in 3DFT acquisitions

SENSE can be applied in both phase-encode directions, thereby increasing the overall reduction factor.

Another feature of SENSE is that the reduction factor can be any value between 1 and the number of coil elements. It is not restricted to integer values. In prescribing a SENSE scan the operator chooses the reconstructed FOV. This must encompass all the signal-producing material otherwise serious artefacts will occur (figure 17.7). In other words SENSE cannot tolerate any inherent aliasing. The calibration procedure is explained further in the box 'To autocalibrate or not'.

17.3.2 mSENSE

Modified SENSE (mSENSE) is a version of SENSE which does not require a separate calibration scan. Instead additional lines are acquired at the centre of k-space

Making SENSE with maths

In the 2 coil example of figure 17.5, the signal at point y is made up of the correct signal for this position and one aliased signal. We can write

$$I_1(y) = C_1(y)S(y) + C_1(y + \Delta Y)S(y + \Delta Y)$$

$$I_2(y) = C_2(y)S(y) + C_2(y + \Delta Y)S(y + \Delta Y)$$

where I_1 and I_2 are the image intensities measured for each coil which have sensitivities C_1 and C_2 . Hence we have two simultaneous equations and two unknowns which can be solved exactly by algebra. The aliasing distance ΔY is

$$\Delta Y = \frac{\text{FOV}_{\text{rec}}}{R}$$

where FOV_{rec} is the reconstruction field of view for PE and R is the SENSE reduction factor. With higher R , we may have two or more aliased signals at point y as well as the correct signal, and we can write

$$I_1(y) = C_1(y)S(y) + C_1(y + \Delta Y)S(y + \Delta Y) + \dots + C_1(y + n_A \Delta Y)S(y + n_A \Delta Y)$$

$$I_2(y) = C_2(y)S(y) + C_2(y + \Delta Y)S(y + \Delta Y) + \dots + C_2(y + n_A \Delta Y)S(y + n_A \Delta Y)$$

where n_A is the number of aliased signals which is spatially variant and depends on the size of the object being imaged. If the object exactly fills the reconstructed FOV, then $n_A = R$ at all points; in all other cases $n_A \leq R$.

We can formalise this in terms of image intensities $I_j(x, y)$ for each of j coil elements:

$$I_j(x, y) = \sum_{n=0}^{n_A} C_j(x, y + n \Delta Y)S(x, y + n \Delta Y)$$

If there are L coil elements, we can write L simultaneous equations as above, most conveniently represented in a matrix equation as

$$\begin{bmatrix} I_1(x, y) \\ I_2(x, y) \\ \vdots \\ I_L(x, y) \end{bmatrix} = \begin{bmatrix} C_1(x, y) & C_1(x, y + \Delta Y) & \dots & C_1(x, y + n_A \Delta Y) \\ C_2(x, y) & C_2(x, y + \Delta Y) & \dots & C_2(x, y + n_A \Delta Y) \\ \vdots & \vdots & \ddots & \vdots \\ C_L(x, y) & C_L(x, y + \Delta Y) & \dots & C_L(x, y + n_A \Delta Y) \end{bmatrix} \begin{bmatrix} S(x, y) \\ S(x, y + \Delta Y) \\ \vdots \\ S(x, y + n_A \Delta Y) \end{bmatrix}$$

which is more economically expressed as

$$\mathbf{I} = \mathbf{C}\mathbf{S}$$

and the solution arises from inverting the $L \times n_A$ element matrix $\mathbf{C}$ to obtain the true image signal $S(x, y)$ for every pixel. This is always possible if there are at least as many coil elements as the maximum number of aliased signals and if the coil sensitivities are sufficiently unique. Notice that since $R > n_A$ for the majority of images, it is possible for R to be greater than the number of elements in the coil. Pre-scan calibration data, or as in the case of mSENSE, reference lines are required to obtain the coil sensitivities.

during the diagnostic scan (figure 17.8). These central lines are extracted for each coil element and used on their own to reconstruct low resolution, unaliased, images from each coil element which may then be used to provide sensitivity maps. A SENSE reconstruction

algorithm can then be used to unfold the images from the sparse k-space data (the blue arrows in figure 17.8). The full time saving by the reduction factor R is not achieved because of the additional lines required for calibration.

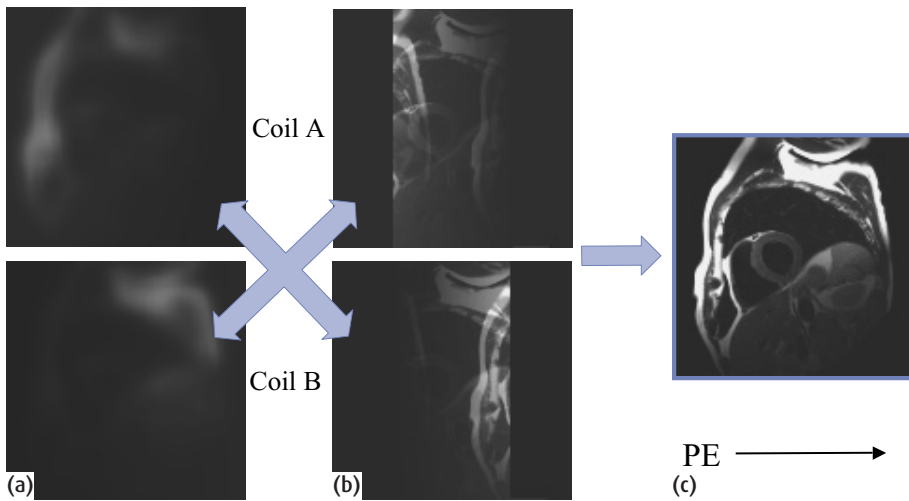


Figure 17.6 Example of SENSE reconstruction steps: (a) reference scans, (b) folded images from each element, (c) reconstructed SENSE image.

17.4 SMASH hits: parallel imaging in k-space

Historically SMASH (SiMultaneous Acquisition of Spatial Harmonics) was the first parallel imaging technique. The term spatial harmonics refers to the spatial frequencies: SMASH works in k-space. It uses combinations of array coil element sensitivities to create virtual phase encoding. In principle it can be applied to any sequence. Just like SENSE it reduces the number of phase-encode gradient steps you need giving a scan time reduction by factor R .

17.4.1 SMASH

Looking back at figure 17.2 we see the signal distributions in the phase-encode direction for various values of k . The principle of SMASH is simply to generate a similar pseudo k-space phase encoding by using the spatial response of the RF reception. In practice this means using arrays of coils and combining the various elements to obtain the required sinusoidal variations in response across space.

Figure 17.9 shows how two simple linear coils could be combined to produce either a uniform response by adding the signals together, or a nonuniform response which resembles a sinusoidal variation by subtraction of the coil sensitivities. In other words by the appropri-

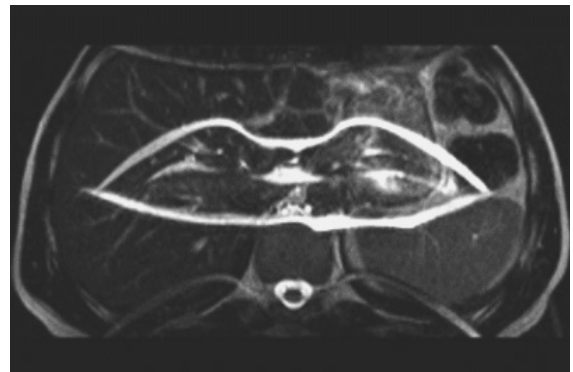


Figure 17.7 SENSE reconstruction "hot lips" artefact resulting from attempting to use SENSE when the unfolded field of view does not contain all of the anatomical region of interest.

ate combination of signals from different coil elements we can obtain a spatial signal distribution which can be made to match either zero spatial frequency (uniform response, $k=0$) or the lowest nonzero spatial frequency ($k=1$). In practice weighted combinations of the coil sensitivities are used. Figure 17.10 shows the coil weights necessary to produce $k=0$ and $k=2$ spatial harmonics for a notional 8 element array coil. Further details are given in the box 'More about spatial harmonics'.

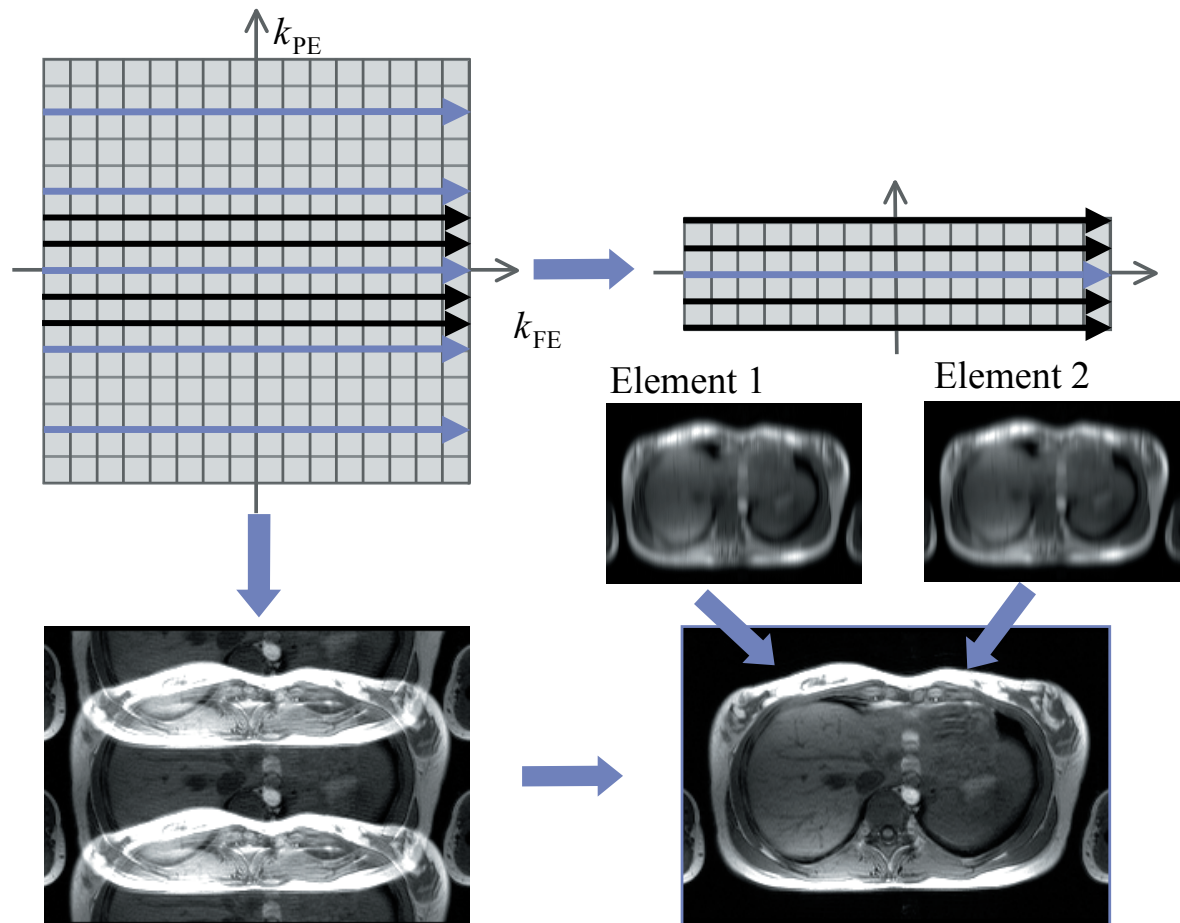


Figure 17.8 Modified SENSE (mSENSE). The centre of k-space is fully sampled and extracted for each coil to form a low resolution image which can be used as a sensitivity map in a SENSE reconstruction. Images courtesy of Siemens Medical Solutions.

This use of RF reception properties to achieve spatial encoding opens the intriguing prospect of an MR acquisition without the use of *any* phase-encode gradients. The RF engineering required to produce a sufficiently complex phased array coil is just too demanding at the moment – the coil would need at least 128 elements spread out along the phase-encode direction! Consequently in SMASH, a combination of phase encoding with gradients, and virtual phase encoding (with array coils) is applied. This is illustrated in figure 17.11 showing the phase change over the FOV for the $k = 2$ PE

gradient from the uniform coil combination ($\Delta k = 0$). By combining the coils differently a modified response ($\Delta k = 1$) is obtained and when added to the effect of the gradient produces a line of virtual phase encoding with $k = 3$. Figure 17.12 shows combinations of gradient and array coil phase encoding to generate six lines of k-space from only three actual acquisitions. In this example then, we can halve the number of phase-encode gradient steps required, and therefore halve the number of excitations required and halve the scan time. The reduction factor R is therefore equal to 2.

To Auto-calibrate or not

In the calibration method used by SENSE low resolution images are acquired from each array element and also from the body coil. The array element images are divided by the body coil image (to remove sensitivity variations due to anatomy rather than coil response). Then various image processing steps (thresholding, filtering, extrapolation, smoothing) are performed to produce the sensitivity map for each coil. Note that due to the thresholding step, the background noise is suppressed in SENSE images.

The calibration scan takes about 20 seconds and once acquired can be used on all subsequent parallel image acquisitions provided the geometry or the patient's position does not change. ASSET (GE) and SPEEDER (Toshiba) also require calibration scans. ASSET does not require the body coil division step.

In mSENSE and GRAPPA (see section 17.4.3) the calibration data is inherent to the main scan acquisition. No body coil image is required and there is no image division. This means that the background noise is not suppressed. Having the calibration (ACS) lines integral to the acquisition also means that there will be no misregistration problem if the patient moves between scans. However, the scan time will be increased by the number of ACS lines used, thus the full reduction factor will not be realised. In GRAPPA, the ACS lines may also be included in the reconstruction, giving a slight increase in SNR.

The acquisition in k-space is illustrated in figure 17.13(a) where we see that k-space is filled completely using a sparser set of phase-encode gradient steps (represented by the blue arrows) and the synthesising of virtual phase-encode lines (the dotted lines) from the combinations of the coils.

17.4.2 Auto-calibrating SMASH

The constraints of coil design make true SMASH very hard to realise on a clinical scanner. In order to make SMASH techniques more applicable to real phased array coils a change of strategy was needed. Instead of making

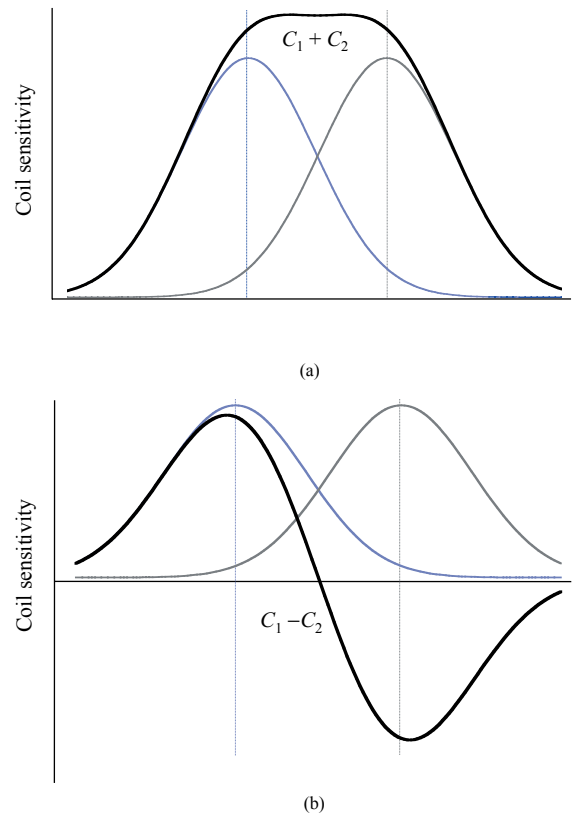


Figure 17.9 Coil sensitivity profile combinations for two loop surface coils, (a) coil combination (addition) to achieve uniform spatial response, (b) coil combination (subtraction) to produce, non-uniform spatial response.

coils to give a 'spatial frequency-type' signal response, why not accept whatever spatial sensitivity response the coils actually give and then work out how to combine their signals to get the desired lines of k-space? The first technique to do this was Auto-SMASH. The principle is shown in figure 17.13(b) where a reduced k-space is acquired by skipping lines (e.g. every third line to give a reduction factor $R=3$). However we also acquire additional 'auto-calibrating signal' (ACS) lines near the centre of k-space. The scanner then computes from the signals acquired from the array coil elements the nearest linear combination of these signals to match the actual acquired ACS line. The coefficients calculated can then be applied throughout the remainder of k-space to fill in

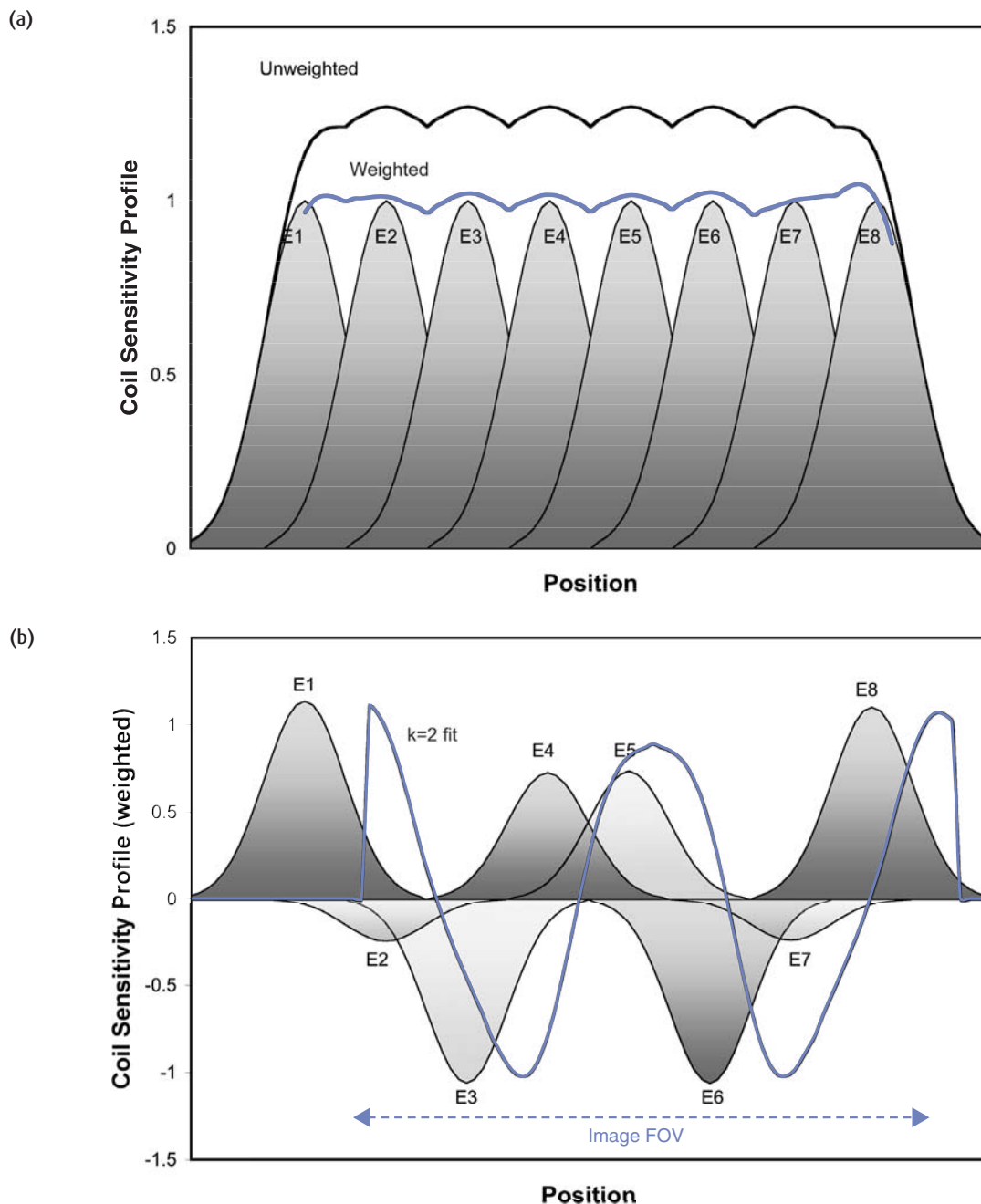
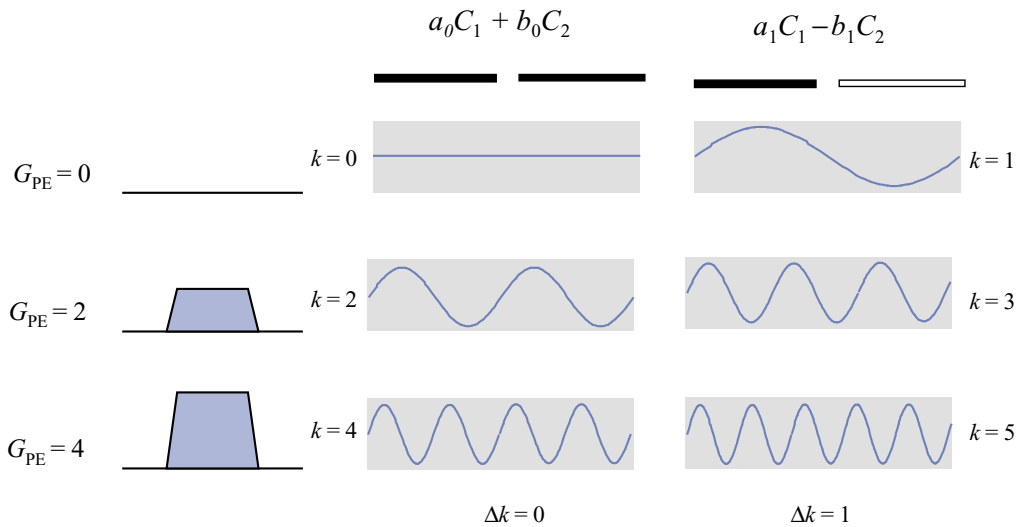
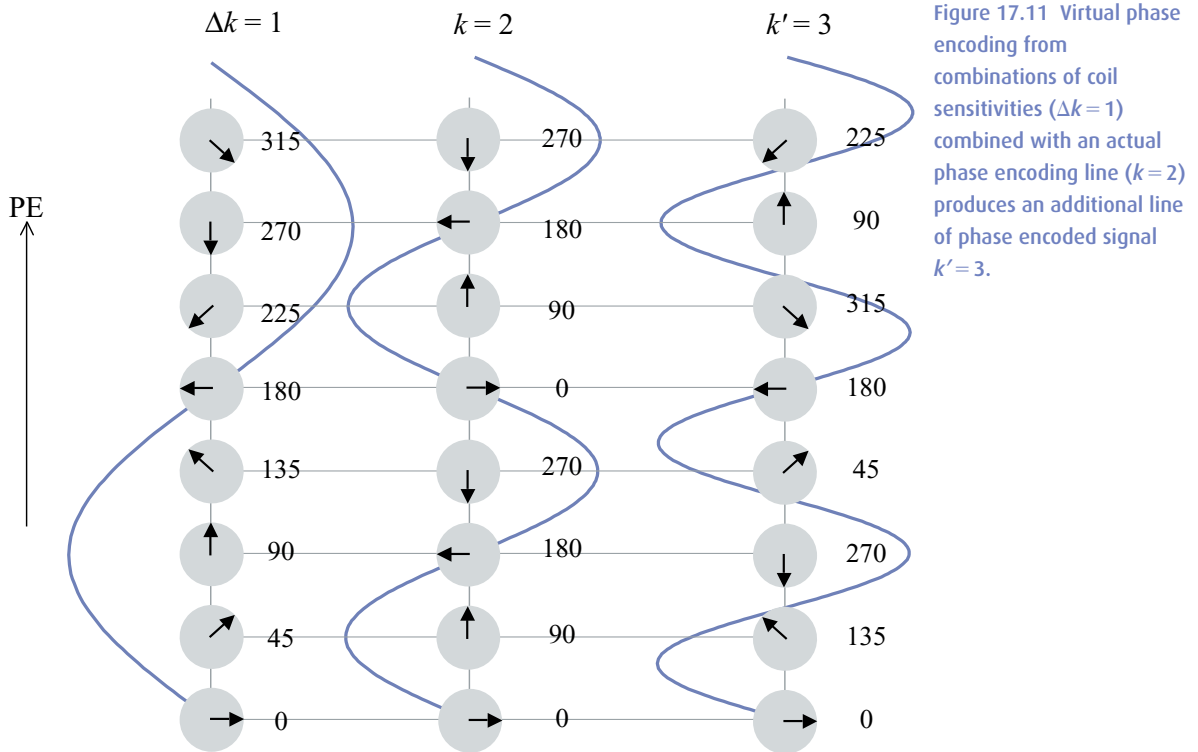


Figure 17.10 (a) Unweighted and optimally weighted coil combinations to produce a uniform ($k = 0$) response for an eight element array coil. (b) Weighted coil sensitivity functions to produce the second harmonic ($k = 2$). Grey areas represent the coil sensitivity profiles multiplied by their respective weights.



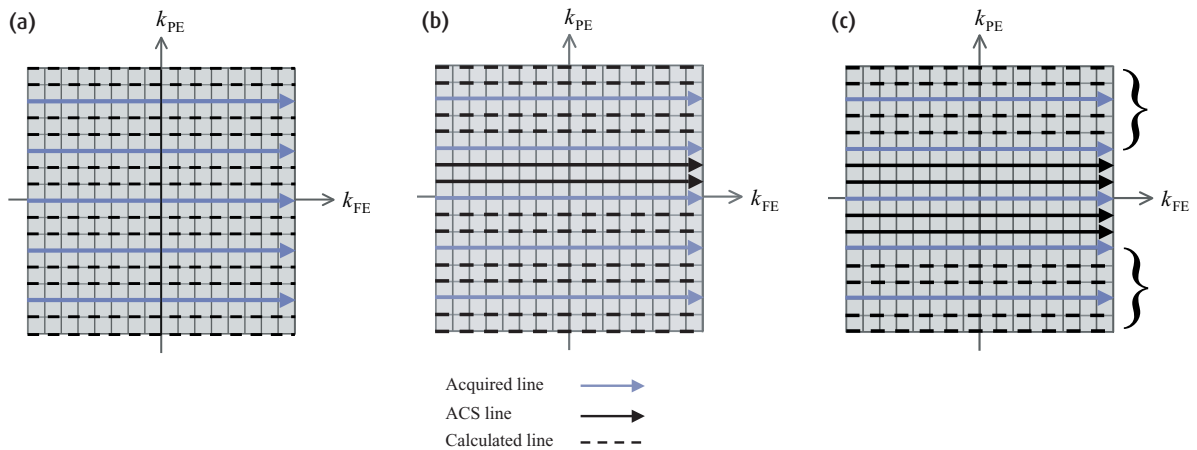


Figure 17.13 k-space scheme for (a) SMASH, (b) Auto-SMASH with 2 ACS lines and (c) VD-Auto-SMASH where an outer reduction factor (ORF) applies away from the centre of k-space. In each example $R = 3$. Arrowed lines are acquired with gradients, the interspersed (non-arrowed) lines are computed from combinations of the coil element signals. The fully filled k-space is Fourier transformed to generate the image.

More about spatial harmonics

The purpose of phase encoding is to produce a linearly varying phase change to the MR signal along the phase-encode axis, as illustrated in the clockface diagram in figure 17.11. In order to generate the rotational nature of phase encoding, we need to add sine and cosine functions (see appendices A2 and A5). Mathematically this is:

$$\exp(i2\pi y k_{PE}) = \cos(2\pi y k_{PE}) + i \sin(2\pi y k_{PE})$$

remembering that k-space uses complex maths. The first harmonics are shown in figure 17.14 for the example of an 8 element coil, along with the $\Delta k = 1$ combination. The figure shows both the calculated harmonics and the ideal ones.

Figure 17.14 also shows the higher harmonic, $\Delta k = 3$, for the same eight element coil, showing that significant deviations exist between the ideal and the fitted harmonics. Errors in the generation of the coefficients will result in image artefacts (see also figure 17.23).

the unacquired lines, before standard reconstruction by Fourier transformation. In practice a minimum of $R-1$ ACS lines are required.

Variable density Auto-SMASH (VD-Auto-SMASH) extends the above principle, but adds additional ACS lines, illustrated in figure 17.13(c). This makes it more robust from artefacts and reconstruction errors. The centre of k-space is more highly sampled and we can talk of an inner and an outer reduction factor. The scan time compared with Auto-SMASH will be longer because of the additional ACS lines. As in Auto-SMASH, a composite k-space (i.e. just one for all the array elements) is acquired and estimated.

17.4.3 GRAPPA

GRAPPA (GeneRALized Autocalibrating Partially Parallel Acquisitions) is a further extension of the Auto-SMASH principle. GRAPPA is illustrated in figure 17.15. It deploys multiple ACS lines and uses data from every coil for the fitting of the appropriate weights for each ACS line. A separate sub-k-space is filled for each array element, which is then reconstructed, giving an image for each element. These separate images are then combined by the usual 'sum-of-squares' algorithm used in

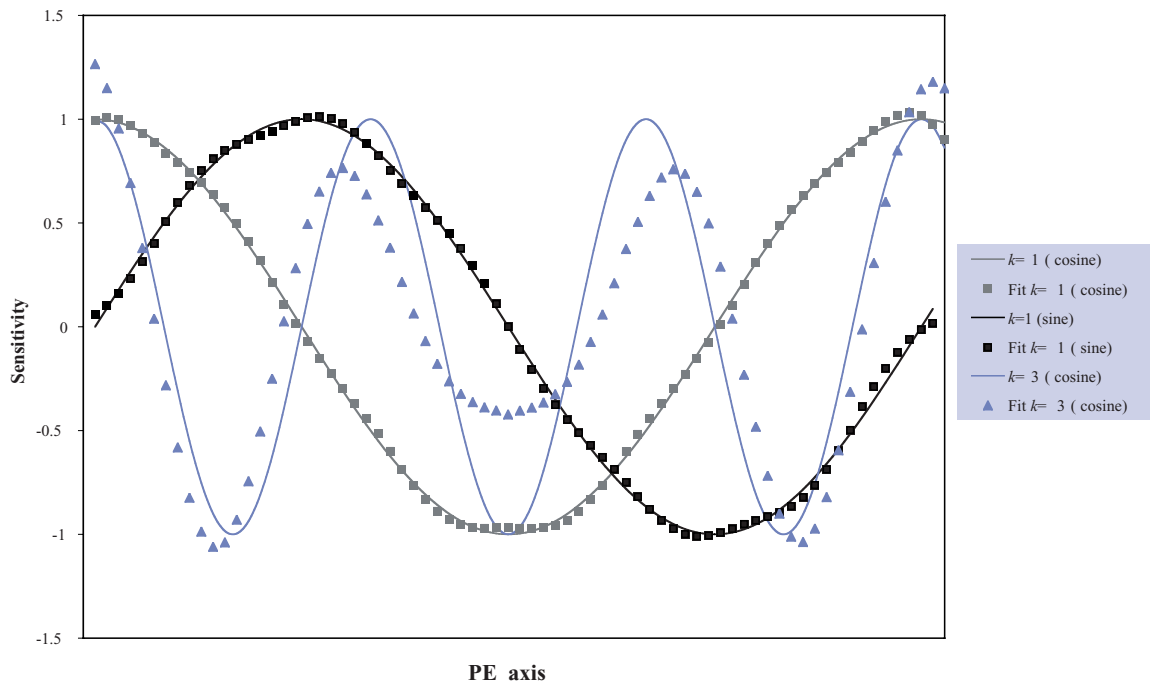


Figure 17.14 Fitted (calculated) and ideal sensitivity profiles for $\Delta k = 1$ (sine and cosine) and cosine profiles for $\Delta k = 3$. Solid lines show the ideal response, points show the weighted fits for an 8-element array.

conventional array coil reconstructions. GRAPPA has been implemented commercially in Siemens scanners where the reduction factor is known as the iPAT factor and parallel imaging (both mSENSE and GRAPPA) goes under the generic name iPAT (where PAT stands for Parallel Acquisition Technique and the 'i' stands for 'integrated' - but we suspect is really just to appear trendy.)

17.5 k-t BLAST (or as we prefer to call her, Katie Blast)

This technique is very new and not yet available on any clinical systems, but it is very promising for dynamic techniques such as cardiac imaging and dynamic contrast enhancement. k-t BLAST (Broad-use Linear Acquisition Speed-up Technique) takes advantage of the fact that most of the image does not change at all during these acquisitions. For example in a typical

short-axis cine of the heart, the chest wall, lungs, spine and muscles do not vary (much) from one frame to another. Once we have the first frame, these voxels do not need to be re-sampled during the rest of the cine. The remaining voxels can therefore be sampled quicker and provide better temporal and/or spatial resolution.

Figure 17.16 illustrates the principle for a single line through the cardiac frame. If we take the Fourier transform of this line through time, we find ourselves in x - f space, which shows the spectrum of motional frequencies. In the middle are all the voxels which do not change through the frames of the cine acquisition: out to the edges are the faster moving voxels which represent the ventricle walls and moving blood. It is obvious that most of this space is empty!

In normal imaging, every point in k -space is sampled at each of t time-points. By skipping some of the k -space points for some of the time-points, k - t space is undersampled. In x - f space, the periodic nature of the signal means that they are folded in. Reconstruction of

SMASH and k-space methods

As usual in MRI, those of a mathematical nature get it easy as the whole thing can be reduced to one equation:

$$S(k_x, k_y) = \iint \rho(x, y) \cdot C(x, y) \cdot \exp\left(\frac{-t}{T_2^*}\right) \cdot \exp(i2\pi x k_{FE}) \cdot \exp(i2\pi y k_{PE}) \cdot dx dy$$

This is similar to the equation for 2DFT given way back in chapter 7, but with one additional term: $C(x, y)$ which represents the coil sensitivities. For an array of coils with sensitivities $C_j(x, y)$

$$\begin{aligned} S(k_x, k_y) &= \iint \rho(x, y) \cdot \left[\sum_j n_j C_j(x, y) \right] \cdot \exp\left(\frac{-t}{T_2^*}\right) \cdot \exp(i2\pi x k_{FE}) \cdot \exp(i2\pi y k_{PE}) \cdot dx dy \\ &= \sum_j n_j \cdot S_j(k_{FE}, k_{PE}) \end{aligned}$$

where n_j are weighting factors for a superposition of coil and signal sensitivities. By judicious choice of weights we can obtain the combinations

$$\sum_j n_j C_j(x, y) = 1 \quad \text{or} \quad \sum_j n'_j C_j(x, y) = \exp(i2\pi y \Delta k_{PE})$$

where Δk is a step in spatial frequency.

Using the first of these yields the standard 2DFT equation for a uniform coil. Using the second gives

$$\begin{aligned} S' &= \iint \rho(x, y) \cdot \exp\left(\frac{-t}{T_2^*}\right) \cdot \exp(i2\pi x k_{FE}) \cdot \exp(i2\pi y [k_{PE} + \Delta k_{PE}]) \cdot dx dy \\ &= S(k_{FE}, k_{PE} + \Delta k_{PE}) \end{aligned}$$

which is the acquired line of k-space offset by Δk . With the appropriate linear combination of coil signals we can therefore acquire purely gradient-phase-encoded signals or we can calculate additional virtual lines of k-space without having to acquire them.

In SMASH the sensitivities are computed from a calibration scan. In Auto-SMASH and GRAPPA additional auto-calibration signal (ACS) lines are acquired and the coil weights calculated from a fit of the ACS data with the known (acquired) neighbouring line or lines:

$$S(k_{FE}, k'_{PE}) = \sum_j n_j S_j(k_{FE}, k_{PE})$$

thus the coefficients n_j can be calculated to give a particular k-space offset.

these signals will give foldover in real space and under-sampling of the temporal frames, just like real-space signals are folded in during parallel imaging. To unfold the signal intensities, k-t BLAST uses a set of training data, usually a low-resolution cine with relatively few frames (figure 17.17). The training data gives an estimation of the signal distribution in x - f space. Combining the training data with the undersampled dynamic data

allows the reconstructor to produce the final cine images.

k-t SENSE is k-t BLAST plus sensitivity encoding. By using the coil sensitivities of a multi-element receive coil, k - t space is undersampled even more, allowing speed-up factors up to 8. The reduced scan time can be traded for more slices (better coverage), higher spatial resolution, better temporal resolution, or a

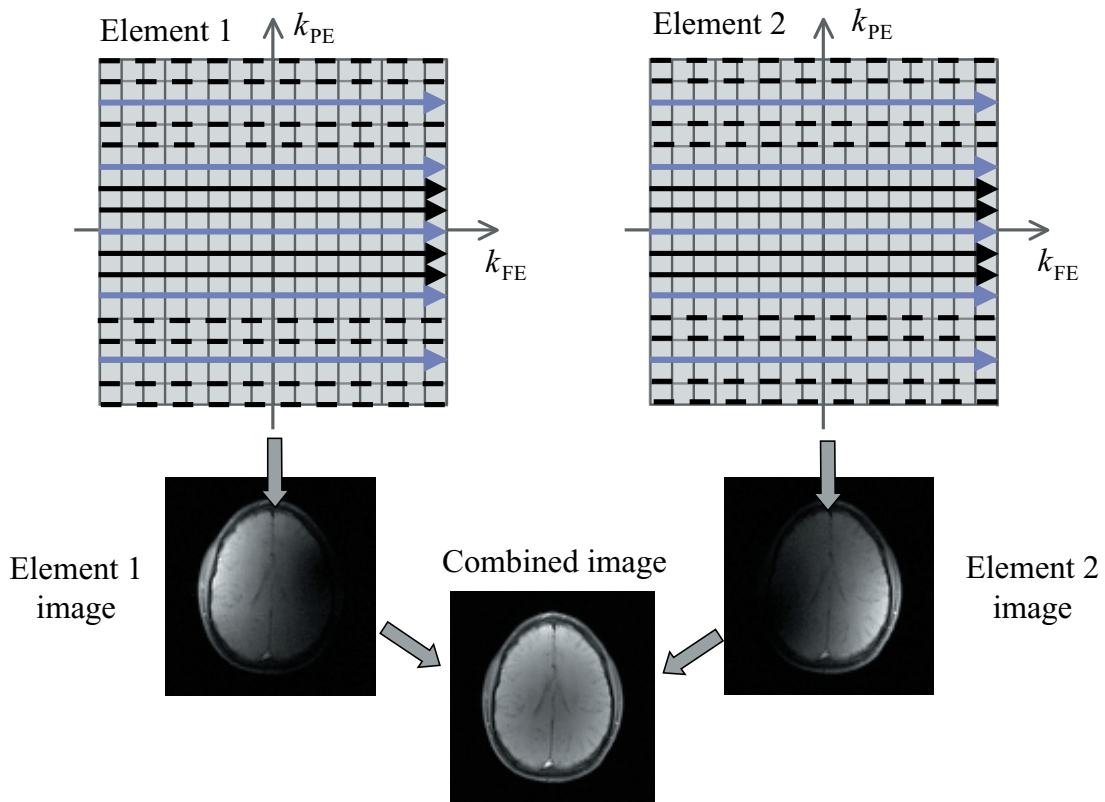


Figure 17.15 GRAPPA acquisition and reconstruction with $R=2$. A separate k-space for each coil element is generated using multiple ACS lines and signal combinations from all the coil elements. Each element's k-space is Fourier transformed to generate images from each element which are then combined as in conventional phased array reconstruction.

combination of all three. For example, velocity mapping through the aorta can be achieved in a 10 s breath-hold instead of a 3 min scan time, or 6 slices may be acquired instead of only one in the same 3 min scan time.

In dynamic contrast enhanced imaging, the anatomy is not changing position but certain voxels will show higher intensity during the injection of gadolinium. In this situation one could acquire a training data set using a small test bolus, but it is preferable to acquire small bits of training data throughout the main bolus injection. k-t BLAST has great potential for applications in dynamic liver imaging, where we need to acquire a large number of slices at a temporal resolution of at least one frame per second.

17.6 Clinical benefits of parallel imaging

The benefits to parallel imaging can be summed up by the 'five Fs': namely faster, further, finer, (more) faithful and fainter (meaning quieter).

Figure 17.18 illustrates how shorter scan times can be achieved with very little discernable loss of image quality. Faster scanning also means shorter breath-holds as shown in figure 17.19 with the potential for reduced respiratory motion artefact or the second F: further, or more slices per breath-hold. The reduction in the number of phase-encode steps acquired means that higher matrices, and hence better resolution, finer images, may be acquired within a reasonable scan time. Thus a 1024 matrix image using $R=2$ can be acquired in

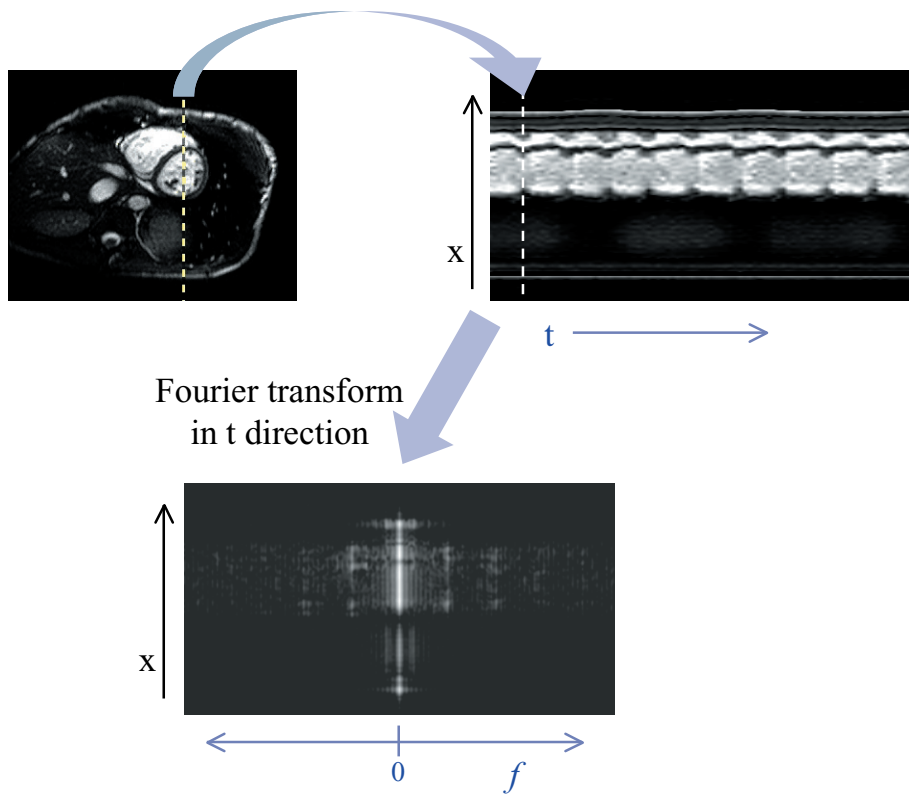


Figure 17.16 In a cine cardiac scan, most of the voxels do not change between frames. By Fourier transforming a line of pixels in the time direction, we can see the frequency distribution of voxel 'motion' (actually it is signal intensity which is changing).

Parallel imaging and high field systems

There is an additional 'F' that we didn't consider earlier which is particularly relevant to high field MR systems: fry. We don't want to overheat our patients! As the SAR increases with the square of field strength (or frequency) high field systems (e.g. 3 T and above) are potentially limited by the permissible SAR. Of course these limits prevent us from frying our patients, but they do result in a need to length TR, or reduce the number of slices or the echo train length (as seen in section 10.2.2). Parallel imaging helps by reducing the number of RF excitations required to form an image, and indeed, without it, even the simplest head scan can prove problematical at 3 T.

the same time as a convention 512 matrix image (figure 17.20).

Parallel imaging may be used to advantage in echo planar acquisitions (EPI) to reduce the effect of susceptibility artefacts (figure 17.21) and to reduce T_2 blurring in single shot fast spin echo or HASTE images (figure 17.22). In addition to reducing the echo train length, it also can enable a shortening of TE. For high field systems, the susceptibility distortions in EPI can be particularly bad. Parallel imaging helps reduce this. Also, particularly for EPI, parallel imaging can help to reduce the acoustic noise.

17.7 Image quality in parallel imaging

We saw in chapter 5 that the image SNR was dependent upon the number of phase-encode lines acquired. This

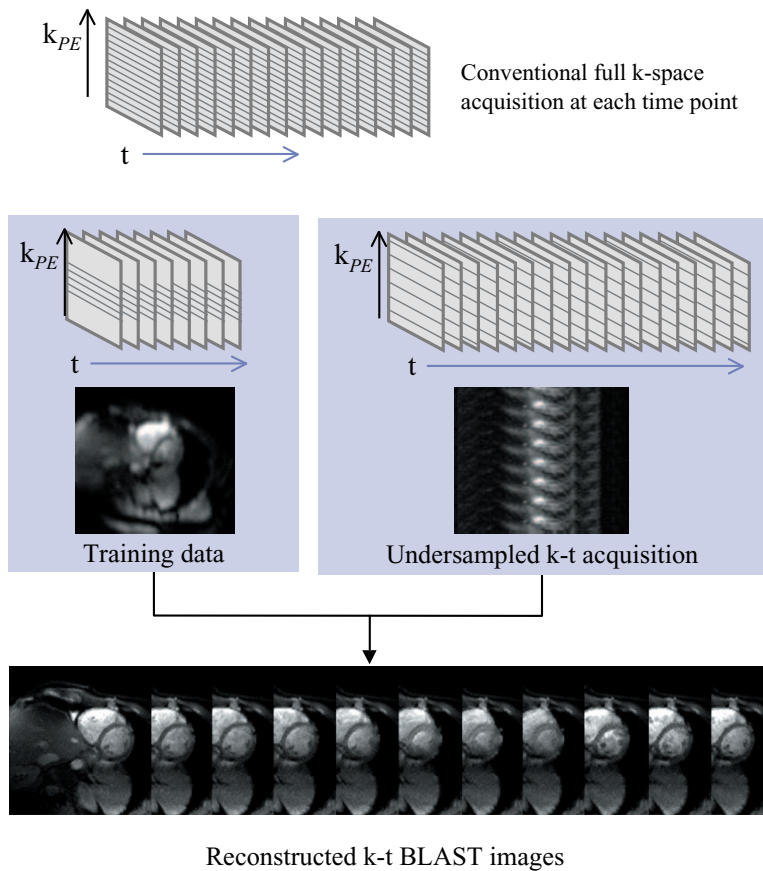


Figure 17.17 A conventional cine scan acquires the whole of k-space at every time point. In k-t BLAST, a set of training data is acquired with low spatial and temporal resolution. This is followed by a rapid acquisition which undersamples in both k-space and time in a well defined pattern. The training data can then be used to unwrap the aliased signals

is also true in parallel imaging. So if we use a reduction factor R , then the SNR will be reduced theoretically by at least $\sqrt{R}$. In practice it may also be reduced by a geometric efficiency factor g which is dependent upon the geometrical arrangement of the array elements and their sensitivities.

If too high a reduction factor is used then artefacts may also appear in the image. These are shown for mSENSE and GRAPPA in figure 17.23. Artefacts also appear if too few ACS lines are used in the auto-calibrating techniques. Typical parallel imaging artefacts are increased noise in the centre of the image (or wherever the coil sensitivities are weakest), and residual foldover ghosts particularly if there is a very strong signal (like fat on T_1 -weighted images).

SNR and SENSE

The SNR in SENSE can be expressed as

$$\text{SNR}_{\text{SENSE}} = \frac{\text{SNR}_{\text{full}}}{g\sqrt{R}}$$

where g is a geometry factor which is always greater than or equal to 1. An additional feature of SNR in SENSE is that the noise is not uniformly distributed across the image as in conventional acquisitions. This can be seen in 17.24 for different R with the level of noise and artefact increasing with R .

The SNR in mSENSE and GRAPPA does not have such a simple relationship with R , as ACS lines can be used in the reconstruction to boost the SNR. Like SENSE, the noise varies spatially across the image in SMASH-type techniques.

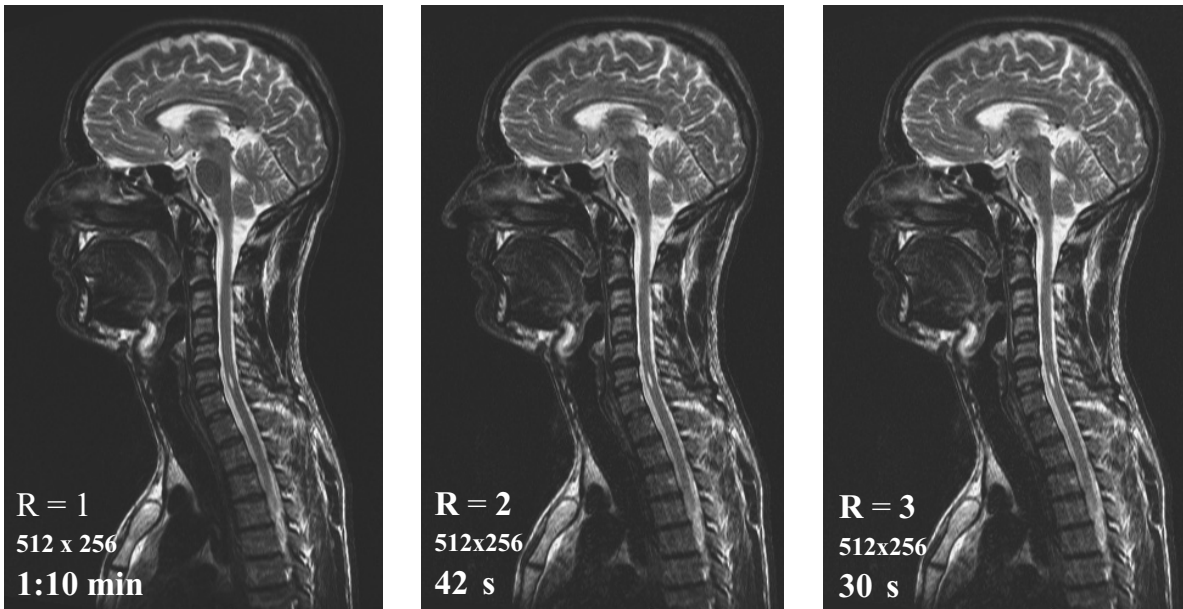


Figure 17.18 Use of parallel imaging (GRAPPA) to speed up T_2 weighted FSE scans. Images courtesy of Siemens Medical Solutions.

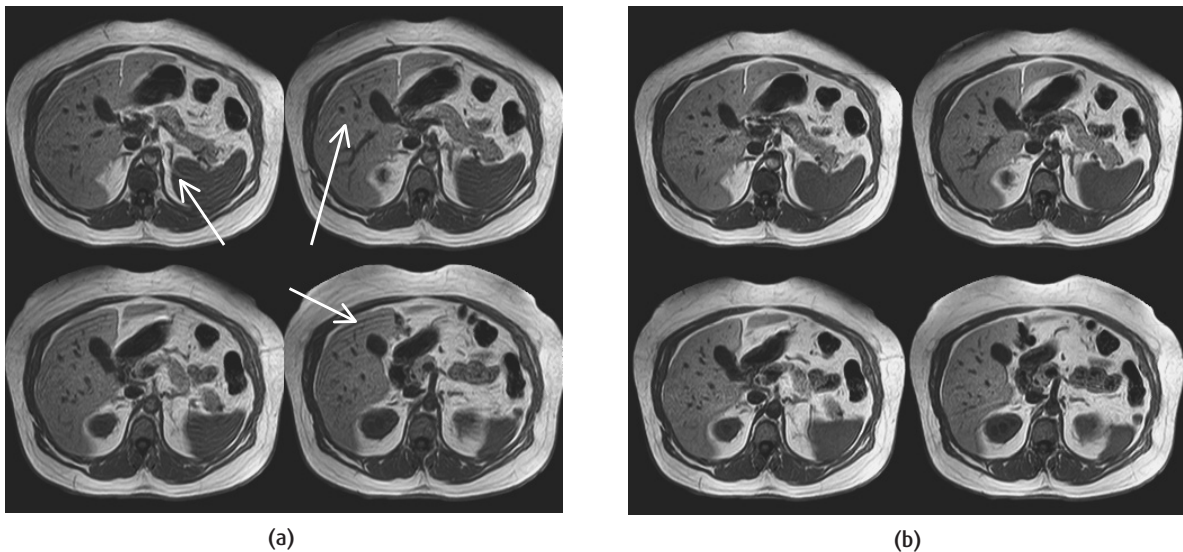


Figure 17.19 Use of SENSE to obtain better slice coverage within a breath-hold, resulting in reduced motional artefact (arrowed), (a) without SENSE: 22 s breath hold (25 slices), (b) with SENSE: 13 s. Images courtesy of Philips Medical Systems.

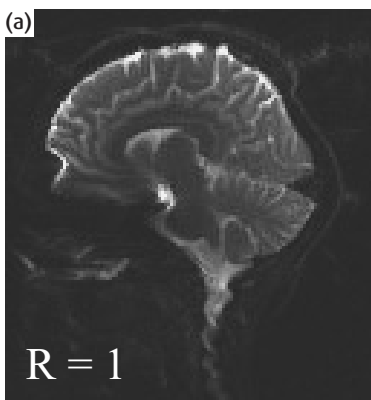


(a)

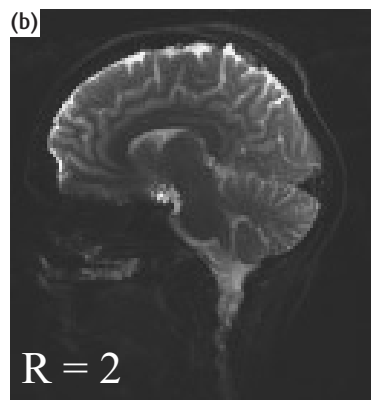


(b)

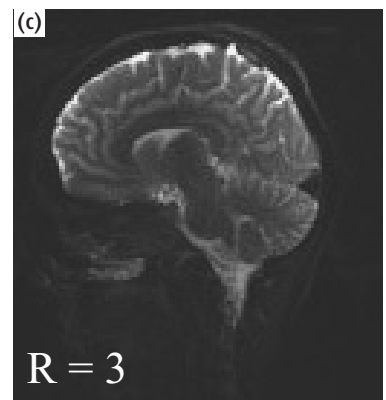
Figure 17.20 Use of parallel imaging (ASSET) to reduce acquisition time for a 1024 matrix acquisitions. Images courtesy of GE Healthcare.



(a)



(b)



(c)

Figure 17.21 Parallel imaging reduces T_2^* distortion in EPI, by shortening the echo train length (increasing the PE bandwidth): (a) Conventional, (b) $R = 2$, (c) $R = 3$. Images courtesy of Siemens Medical Solutions.

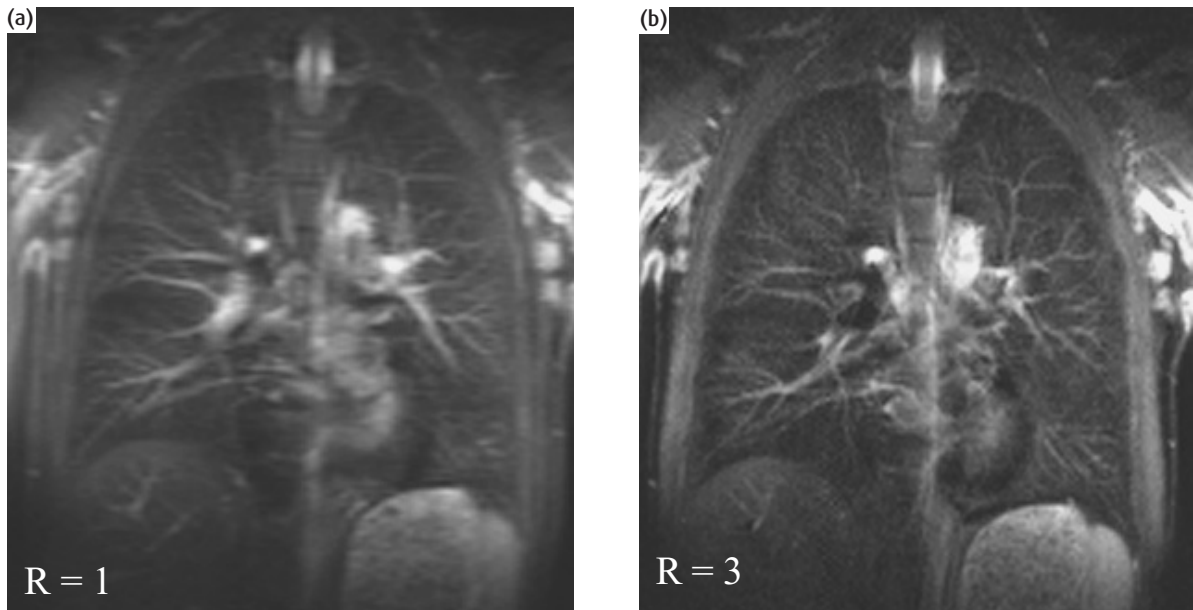


Figure 17.22 Reduction of T_2 blurring in HASTE imaging of the lungs. (a) Conventional 128×256 (207 ms), echo spacing 2.88 ms, (b) 256×256 (149 ms), effective echo spacing 0.96 ms. Images courtesy of Siemens Medical Solutions.

Parallel EPI

Figure 17.25 shows a reduced k-space trajectory for EPI and the related gradient scheme with and without reduction. The shorter echo train results in a higher PE bandwidth and hence less susceptibility distortion artefacts and reduces the minimum TE. The single shot nature of EPI however has implications for the reference images and calibration procedure. For SENSE and similar techniques, a standard calibration is run prior to the EPI acquisition(s) as for other sequences. For GRAPPA, the autocalibration traversal of the centre of k-space takes place before the reduced acquisition. In a repeated series of acquisitions as required for fMRI, this results in one additional 'TR' period. Another odd thing about EPI-GRAPPA is that, because the ACS data is real data and can be used in the reconstruction, the SNR for a reduction factor of 2 is higher than without parallel imaging, even if TE is kept the same. For $R=3$ and above SNR and image quality is degraded as the fitting to the ACS lines becomes more approximate.

We have already noted that for a SENSE acquisition, the field of view must be sufficiently large to encompass all the signal producing regions. If it is not, then the images cannot be unfolded properly.

17.8 Non-Cartesian acquisition schemes

Projection reconstruction (PR) was the first MR k-space trajectory used by Lauterbur to produce the first MR image of two tubes of water. This method is similar to the reconstruction of computerised tomography (CT) images. In CT, projections of the object are acquired at a number of angles around the object. The images are then reconstructed by 'back-projecting' the individual projections. Since simple back projection results in blurred images the projection data is usually filtered before reconstruction. In the earliest MR scanners magnetic field inhomogeneity and gradient nonlinearities resulted in excessive image blurring (see figure 1.3) and PR was subsequently replaced by the 'spin-warp' (2DFT) technique which was much more forgiving (see chapter 7). However with technical improvements in

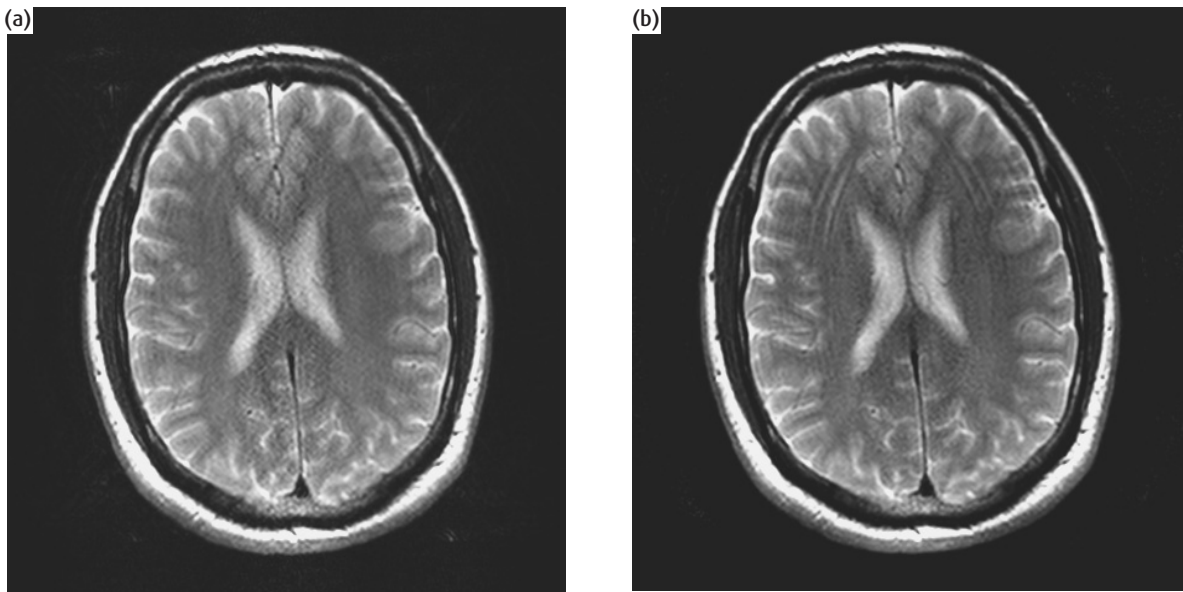


Figure 17.23 Parallel imaging artefacts at high reduction factor ($R=3$) with 8 element head array coil: (a) GRAPPA (b) mSENSE.

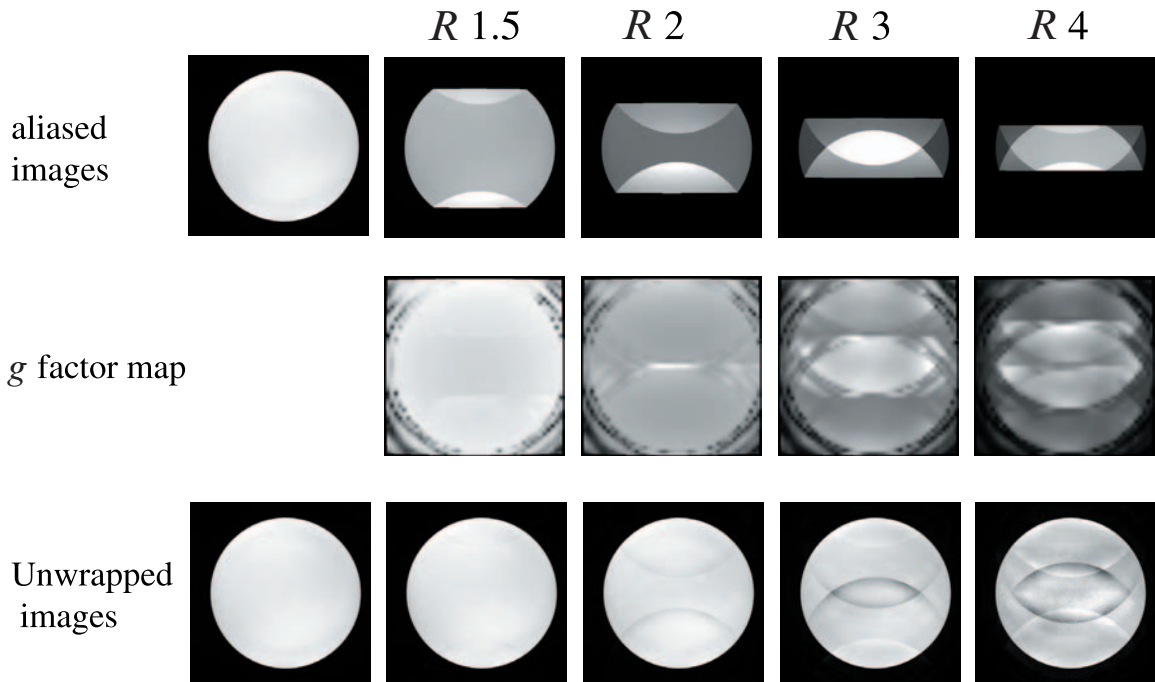
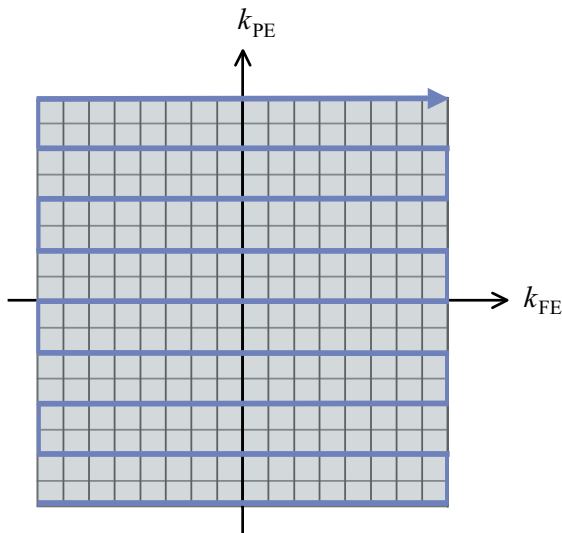


Figure 17.24 Image noise and reconstruction artefacts in SENSE at various reduction factors. Image courtesy of Philips Medical Systems.

scanner performance PR has been reborn with a new name: radial imaging.

17.8.1 2D radial imaging

In radial imaging we acquire a number of 2D projections through the object at different angles, ϕ . To



produce the radial projection pattern frequency encoding is applied simultaneously on two physical axes (e.g. x and y for transverse slices) with varying amplitudes to produce the rotational pattern. The pulse sequence simply involves slice selection and frequency encoding; there is no concept of a phase encoding direction. Figure 17.26(a) shows a 2D gradient echo radial sequence and twelve radial projections overlaid on a Cartesian k-space. In figure 17.26(b) the sequence is shown for the projection at $\phi = 45^\circ$ i.e. the G_x and G_y gradients have the same amplitude. In radial imaging the data is no longer reconstructed using filtered back projection but is 'regridded' onto a conventional Cartesian (x, y) k-space and then reconstructed via direct Fourier transformation (see box 'Re: gridding').

17.8.2 3D radial imaging

The radial approach can also be extended into 3D by acquiring radial projections in 3D, i.e. spherical k-space coverage, this has sometimes been termed the Koosh ball trajectory due to its similarity with the child's toy and adult stress-buster! The proper name for this method is VIPR (Vastly Undersampled Isotropic

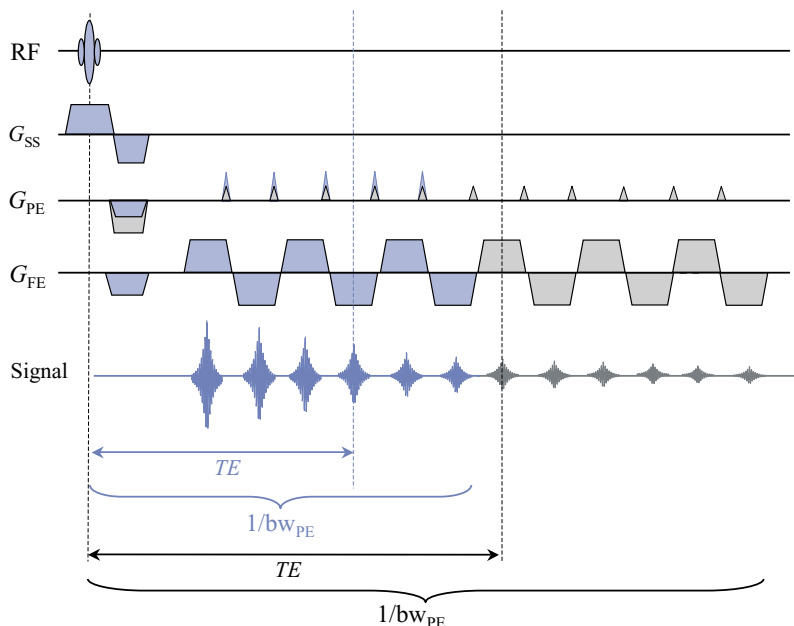
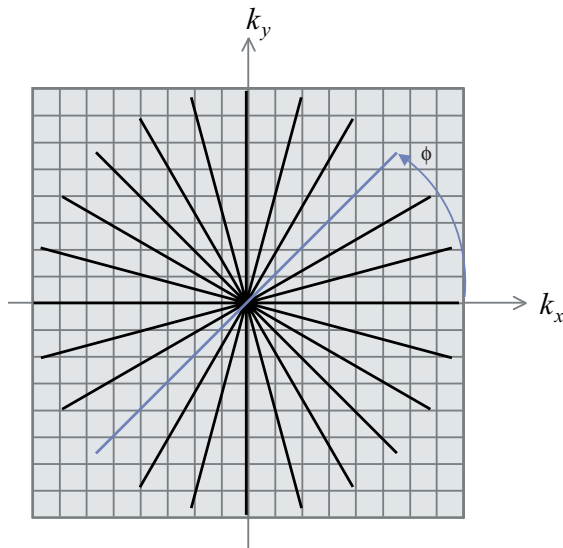


Figure 17.25 Parallel imaging and EPI: (a) k-space path of EPI with $R = 2$. (b) EPI sequence corresponding to k-space in (a) with (blue) and without (grey) a reduction factor of 2. The use of parallel imaging reduces TE and increases the PE bandwidth, resulting in less susceptibility distortion.

Projection Reconstruction). A VIPR acquisition of 27 000 projections can result in a 3D volume with an isotropic resolution of $256 \times 256 \times 256$ being acquired every few seconds, ideal for dynamic 3D contrast enhanced MRA studies.



Re: Gridding

In conventional MRI raw data are acquired along a uniformly sampled rectilinear k-space trajectory and the image reconstructed by direct Fourier transformation. In radial and spiral imaging the k-space trajectories are nonuniform. Whilst any nonuniform data can be reconstructed using extensions of the standard discrete Fourier transform (DFT) algorithm the methods are generally far too slow to be clinically useable. Instead, in a process called 'regridding', the data is resampled or interpolated onto a uniform rectilinear grid prior to a standard FFT. Figure 17.27 shows the principle of gridding in a very simple way where each acquired data point contributes to the signal in four nearest neighbours on the Cartesian grid. An important part of gridding is that the data also need to be corrected for the nonuniform sampling density, i.e. some sequences like radial and spiral imaging heavily oversample the centre of k-space. Each data acquisition point therefore needs to be multiplied by an appropriate 'weighting' factor to compensate for this effect, usually based on a Bessel function.

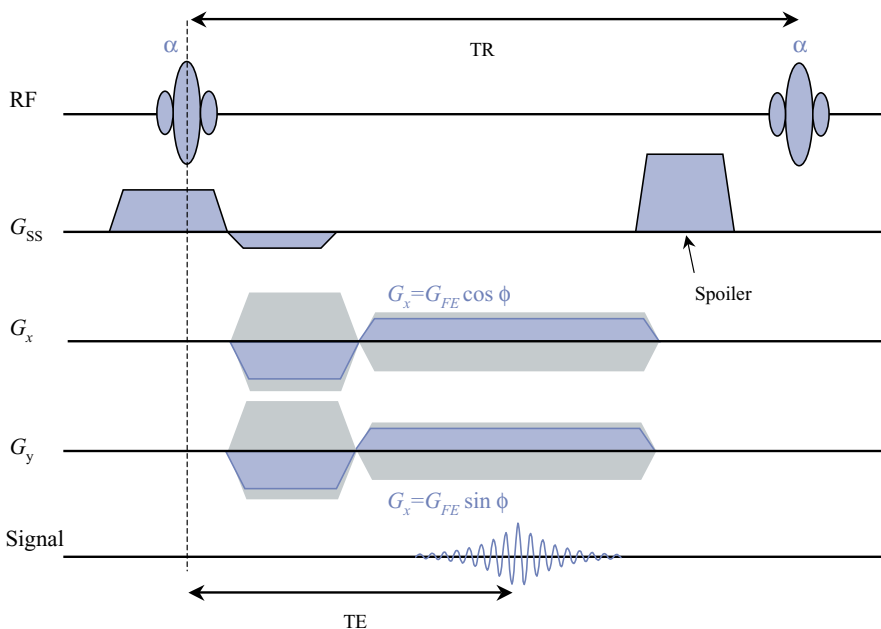


Figure 17.26 Radial scanning: (a) k-space for radial acquisition, the projection for $\phi = 45^\circ$ shown in blue. Regridding to the underlying Cartesian k-space is required prior to reconstruction. (b) Pulse sequence for radial acquisition for $\phi = 45^\circ$. There is no phase encoding and the amplitude of G_x and G_y vary to produce the radii, represented by the grey areas.

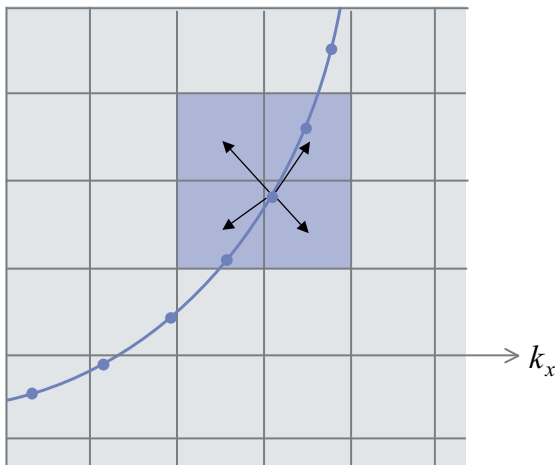


Figure 17.27 Regridding: each point affects four nearest neighbours. The blue line and points represent actual k-space trajectory and sampled points.

17.8.3 Applications of radial imaging: ultra short TE

Radial imaging has a number of advantages. Firstly because no phase encoding gradient is used the minimum TE for gradient echo based radial imaging can be made very short. Using nonselective or half-pulse excitation together with half-Fourier readout and ramped sampling TE can be as short as a few hundred microseconds. Furthermore if hardware changes are made to the system such that system delays and switching times are substantially reduced then echo times as short as $8\mu\text{s}$ can be achieved. This makes it possible to image previously unobservable tissues such as cortical bone, tendons, ligaments and menisci which have very short T_2 (figure 17.28). The term **Ultra-short TE** is used to describe these applications and it gives us another new acronym, UTE. The UTE images themselves are heavily PD weighted, so it is common to acquire a second echo at a later time (e.g. 4 ms) and subtract it from the UTE image. The subtracted image highlights the short T_2 tissues as bright structures against a darker background.

Secondly, because each radial projection passes through the centre of k-space the signals are effectively heavily averaged. This means that motion artefacts are

Sampling in radial imaging

The oversampling of the centre of k-space means that radial acquisitions take $\pi/2$ times longer than conventional Cartesian acquisitions for the same matrix. The scan time can be reduced by undersampling the radial data, i.e. decreasing the number of spokes. In conventional Cartesian imaging this results in image aliasing. In radial imaging it results in radially positioned streak artefacts (figure 17.29). In the same way as for undersampled Cartesian imaging, reducing the number of radial spokes does not compromise spatial resolution but reduces SNR and leads to increased streaking artefacts. However, with a high enough degree of undersampling the artefacts become a nearly uniform background haze instead of discrete streaks. With an appropriate trade-off between acquisition time and artefact then undersampled PR offers a means of rapidly acquiring high resolution images.

considerably reduced but at the expense of increased blurring.

17.8.4. PROPELLER

PROPELLER (**P**eriodically **R**otated **O**verlapping **P**arallel **L**ines with **E**nhanced **R**econstruction) is a hybrid Cartesian/radial acquisition. PROPELLER acquires a number of parallel lines, each rotated about its centre through a series of angles in order to fully sample k-space (figure 17.30). The k-space path then resembles an aircraft propeller, the 'blades' being the rotating block of parallel lines. If each blade only comprised one line of raw data then the technique would be identical to a conventional radial acquisition.

The most common implementation of PROPELLER uses a fast spin echo (FSE) train to acquire each blade. Like radial imaging PROPELLER takes $\pi/2$ times longer than a conventional Cartesian FSE acquisition, e.g. in order to match a conventional acquisition with an N_{PE} of 480 lines, using a PROPELLER acquisition with an FSE ETL of 28, would require $(480 \div 28) \times \pi/2 \approx 27$ blades. The scan time would therefore be $\text{TR} \times 27$.

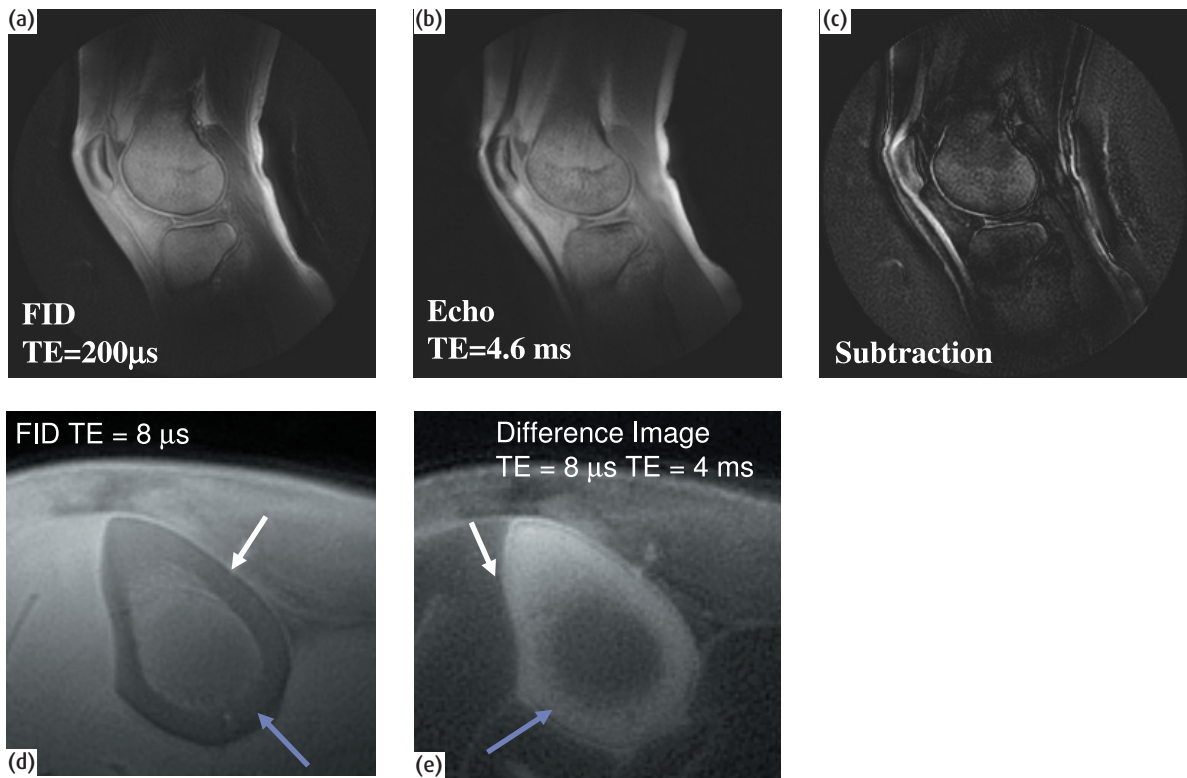


Figure 17.28 (Top) Ultra-short TE images of normal knee: FID, gradient echo and subtracted image which highlights the short T_2 components such as the femoro-patellar tendon. Images courtesy of Philips Medical Systems. (Bottom) Normal tibia: FID and subtracted image, showing high signal in cortical bone ($T_2 = 500 \mu\text{s}$, blue arrows) and periosteum ($T_2 = 5\text{--}11 \text{ ms}$, white arrows). Images courtesy of Jean Brittain, GE Healthcare Technologies.

The advantage of using PROPELLER is that the centre of k-space is heavily oversampled reducing any artefacts due to patient motion. Furthermore, since each blade samples the centre of k-space any rotational or translational motion between blades during the PROPELLER reconstruction can be identified by any inconsistencies between each blade. Depending upon the similarity between blades data can either be corrected to account for the motion or individual blades may even be discarded. PROPELLER can therefore produce good quality images even in the presence of quite severe motion. Figure 17.31 shows two T_2 -weighted FSE acquisitions in a normal volunteer who was asked to rotate their head from side to side during a conventional FSE acquisition and a PROPELLER acquisition.

PROPELLER can also be used with diffusion weighting. Although PROPELLER is significantly slower than EPI, the use of an FSE readout means that susceptibility issues are significantly reduced and that the overall image SNR is much higher. The issue of patient motion during the extended acquisition time is addressed by the inherent PROPELLER motion correction.

17.8.5 Spiral

In spiral imaging the k-space trajectory samples data along an Archimedean spiral (see box 'k-space spirograph'). Spiral acquisitions can achieve greater scan efficiency than conventional Cartesian acquisitions, i.e.



Figure 17.29 Radial artefacts: streak artefacts (very similar to CT) can be seen in the background.

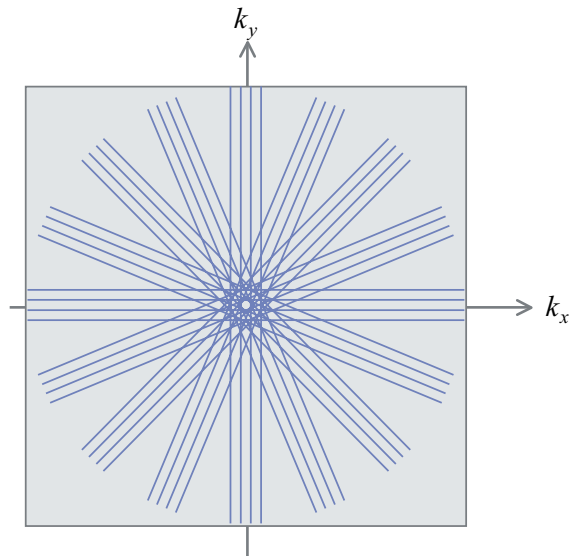


Figure 17.30 k-space trajectories for PROPELLER acquisition with 4 echoes per blade and 8 blades.

k-space spirograph

Spiral trajectories use an Archimedean spiral which is defined by

$$r = a\theta$$

where r is the radius, a is a constant and θ the angle.

The k-space path for a simple constant spacing spiral is given by

$$k_x = \frac{N_{\text{shot}}}{2\pi \cdot \text{FOV}} \theta \sin \theta$$

$$k_y = \frac{N_{\text{shot}}}{2\pi \cdot \text{FOV}} \theta \cos \theta$$

where N_{shot} is the number of interleaved spirals or shots. The gradient waveforms necessary to produce a spiral k-space trajectory come from the derivative of k with respect to time:

$$G_x = \frac{N_{\text{shot}}}{\gamma \cdot \text{FOV}} \cdot \frac{d\theta}{dt} (\sin \theta + \theta \cos \theta)$$

$$G_y = \frac{N_{\text{shot}}}{\gamma \cdot \text{FOV}} \cdot \frac{d\theta}{dt} (\cos \theta - \theta \sin \theta)$$

These are shown in figure 17.32a. In practice slew rate constraints need to be considered and the rotation velocity $d\theta/dt$ may not be constant.

a spiral acquisition can cover a greater portion of k-space with each RF excitation. As the spiral readout starts at the centre of k-space spiral sequences can have very short echo times. Spiral acquisitions may be either single-shot or multi-shot. In a single-shot acquisition all of the k-space data is acquired following a single RF excitation. Figure 17.32 shows the k-space trajectory for a single shot acquisition and the in-plane gradient waveforms required to generate this trajectory. Like radial imaging there is no longer any concept of phase or frequency encoding and data is constantly acquired during the spiral readout.

For multi-shot acquisitions a number (N_{shot}) of spiral interleaves are performed, each shot being rotated by an angle of $\pm 2\pi/N_{\text{shot}}$ (figure 17.33). Figure 17.34 shows a phantom image acquired with a single shot spiral comprising 16384 complex data points, and a multi-shot with 16 interleaves each of 4096 data points.

Since spirals are a method of in-plane data readout then they can be combined with any other pulse sequences, for example spin echo. The spirals can also be acquired in a reverse fashion so that the centre of

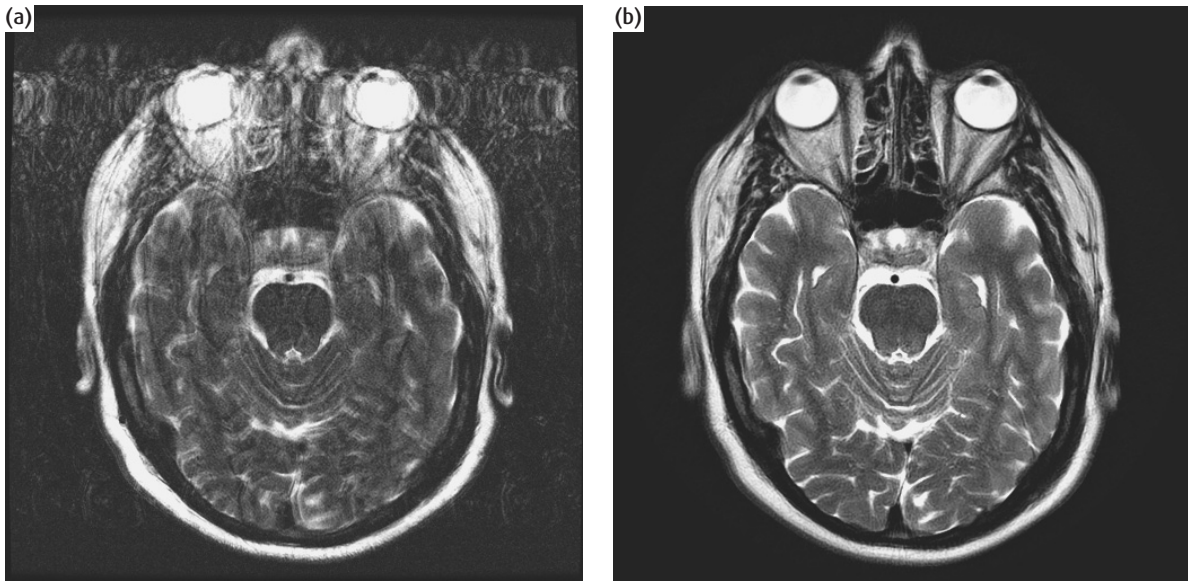


Figure 17.31 Effect of head motion on (a) convention FSE (scan time 2 min 41 s) and (b) PROPELLER (scan time 3 min 12 s).

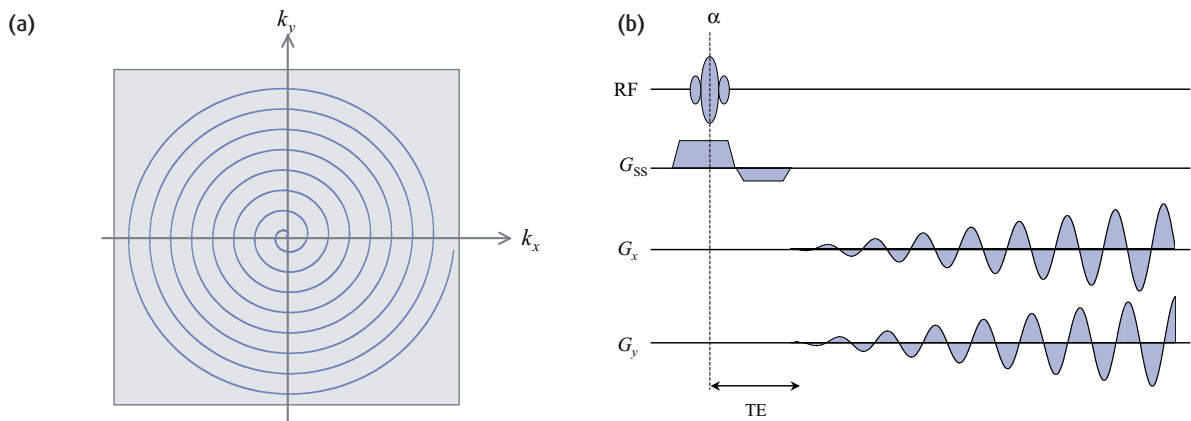


Figure 17.32 k-space for single shot spiral acquisition, (b) spiral pulse sequence to generate the k-space trajectory in (a).

k-space is acquired last. This is one way of acquiring a T_2^* -weighted spiral acquisition suitable for fMRI studies. Spirals can also be extended to 3D, the simplest

method being to incorporate a conventional slice select phase encoding, the resultant data is often called a stack of spirals.

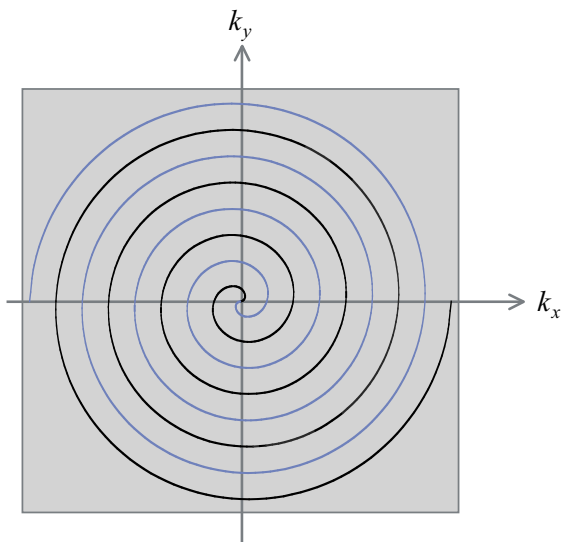


Figure 17.33 Multi-shot spiral k-space with 2 interleaved shots.

17.9. Epilogue: the final frontier

To quote the old human rights song, we've come a long, long way – both since the genesis of modern MRI and since the start of the book. We started with an allusion to rocket science. In this last section we delve into what may turn out to be science fiction, or mere star-gazing. What will tomorrow's MR scanner be like, and what will it do – or possibly, what won't it do? Expect the following:

- The use of interventional MRI will increase and expand, coupled with a proliferation of minimally invasive therapies, including gene therapies.
- Interventional MR systems may use high-temperature superconducting flat magnets (like fridge

Problems with spirals

The major problem with spiral acquisitions is the blurring due to off resonant spins. These frequency offsets can come from B_0 field inhomogeneity, local susceptibility effects and chemical shifts. In conventional Cartesian imaging these frequency offsets result in a simple shift in the frequency encoding direction, e.g. chemical shift artefact, whilst in spiral imaging, since the trajectory is changing simultaneously in both in-plane directions the effect is a 2D blurring (figure 34(c)). To reduce this blurring spiral imaging sequences are usually performed with a water-only excitation (spatial-spectral) excitation (Chapter 12). To correct for the effects due to B_0 inhomogeneity, it is usual for the system to rapidly acquire a field map to determine the frequency offset as a function of spatial location. This can easily be done during the prescan period by acquiring two single shot spiral images with slightly different TEs and constructing a phase map by complex subtraction. The calculated frequency shifts can then be incorporated into the spiral gridding algorithm to correct for the off-resonance spins.

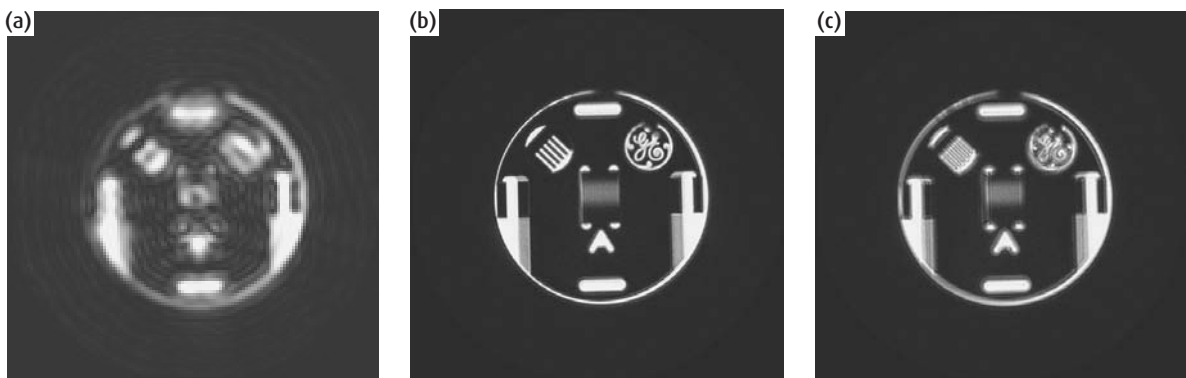


Figure 17.34 (a) Single shot spiral image (b) 16 shot interleaved spiral image (c) 16 shot interleaved spiral but with 50 Hz offset.

magnets) incorporated into the operating room couch.

- For screening we may see walk-through magnets.
- New extensions to parallel imaging will render EPI unnecessary except in some functional neurological examinations (i.e. just like the present!).
- Field strengths will of course keep increasing, and 3T will become the norm for state-of-the-art clinical scanning.
- There will be new 'intelligent' contrast agents for molecular-specific proton imaging.
- Finally spectroscopy will be deployed as a routine clinical application.

See also:

Section 7.7.2 Totally phased: phase encoding

Section 9.4.6 Phased array coils

FURTHER READING

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